

Lecture 6: Diagnostics I (model adequacy checking)

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1 Diagnostics for residuals (residual analysis)

- Scaling residuals
- Diagnostic plots of residuals
 - Aided variable plot

2 Overview of tests involving residuals

- Tests for constancy of error variance
 - Breusch-Pagan test

The major assumptions that we have made thus far in our study of regression analysis are as follows:

- 1 The relationship between the response Y and the regressors X_i is linear, at least approximately.
- 2 The error term ϵ has zero mean.
- 3 The error term ϵ has constant variance σ^2 .
- 4 The errors are uncorrelated.
- 5 The errors are normally distributed.

In this lecture we present several methods useful for diagnosing violations of the basic regression assumptions.

1 Diagnostics for residuals (residual analysis)

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Properties of residuals

- Recall: residual e_i is the difference between the observed value Y_i and the fitted value $\hat{Y}_i$.
- Residual can be regarded as the observed error in distinction to the unknown true error $\epsilon_i = Y_i - E\{Y_i\}$.
- ϵ_i are assumed independent normal random variables $N(0, \sigma^2)$.
- Basic idea of **residual analysis** is verifying if the observed residuals e_i reflect the properties assumed for the ϵ_i .

Properties of residuals

- Mean $\bar{e} = \frac{\sum e_i}{n} = 0$
- Variance (for sample)

$$s^2 = \frac{\sum (e_i - \bar{e})^2}{n - p} = \frac{\sum e_i^2}{n - p} = \frac{SSE}{n - p} = MSE$$

If the model is appropriate, MSE is an unbiased estimator of the variance of the error terms σ^2 .

- Non-independence feature of the residuals:
 - The residuals e_i are actually not independent random variables because they involve the fitted values $\hat{Y}_i$ which are based on the same fitted regression function.
 - However, when the sample size is large in comparison to the number of parameters in the regression model, the dependency effect among the residuals e_i is relatively unimportant and can be ignored.

Scaled residuals: Standardized Residuals (a.k.a. semistudentized residuals)

- a.k.a. semistudentized residuals
- Since MSE is the unbiased estimator of σ , it is natural to consider the following form of standardization

$$e_i^* = \frac{e_i - \bar{e}}{\sqrt{MSE}} = \frac{e_i}{\sqrt{MSE}}$$

- The standardized residuals have mean zero and approximately unit variance. Consequently, a large standardized residual ($e_i^* > 3$, say) potentially indicates an outlier.

Scaled residuals: Studentized Residuals

We can improve the residual scaling by dividing e_i by the exact standard deviation of the i -th residual.

- **Studentized residual** is the ratio of e_i to $s\{e_i\}$

$$r_i = \frac{e_i}{s\{e_i\}}$$

where $s\{e_i\}$ is an estimator of the standard deviation of residuals

$$s\{e_i\} = \sqrt{MSE(1 - h_{ii})}.$$

- Since any point with a large residual and a large h_{ii} is potentially highly influential on the least-squares fit, examination of the studentized residuals is generally recommended.
- If n is large, studentized residuals may not differ dramatically from standardized residuals.

Scaled residuals: Deleted Residuals (a.k.a. PRESS residuals)

- $\hat{Y}_{i(i)}$: delete the i th case, fit the regression function to the remaining $n - 1$ cases, and obtain the point estimate at X_i .
- Deleted residual for the i th case is defined as $d_i = Y_i - \hat{Y}_{i(i)}$.
- It can be proved that

$$d_i = \frac{e_i}{1 - h_{ii}}$$

- $h_{ii} \uparrow$ implies $\frac{d_i}{e_i} \uparrow$.
- Residuals associated with points for which h_{ii} is large will have large PRESS residuals. These points will generally be **high influence** points.

Scaled residuals: Studentized deleted residuals

- Combining studentized residual and deleted residual, we have **studentized deleted residual**, which is defined as follow:

$$t_i = \frac{d_i}{s\{d_i\}}$$

- a.k.a externally studentized residual, or R-student
- The studentized deleted residuals t_i can be calculated from the residuals e_i , the error sum of squares SSE , and the hat matrix values h_{ii} as follows

$$t_i = e_i \left[\frac{n - p - 1}{SSE(1 - h_{ii}) - e_i^2} \right]^{1/2}$$

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Diagnostic plots of residuals

- Plot of residuals against predictor variable.
- Plot of absolute or squared residuals against predictor variable.
- Plot of residuals against fitted values.
- Plot of residuals against time or other sequence.
- Plots of residuals against omitted predictor variables.
- Box plot of residuals.
- Normal probability plot of residuals.

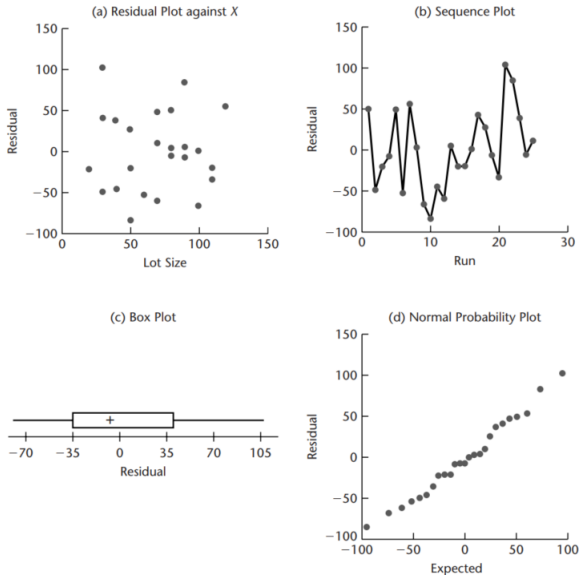


Figure: Diagnostic residual plots, Toluca company example.

Transit example

- A study of the relation between *maps distributed* and *bus ridership* in eight test cities.
 - X is the number of bus transit maps distributed free to residents
 - Y is the increase during the test period in average daily bus ridership during nonpeak hours.

City i	(1) Increase in Ridership (thousands) Y_i	(2) Maps Distributed (thousands) X_i	(3) Fitted Value $\hat{Y}_i$	(4) Residual $Y_i - \hat{Y}_i = e_i$
1	.60	80	1.66	-1.06
2	6.70	220	7.75	-1.05
3	5.30	140	4.27	1.03
4	4.00	120	3.40	.60
5	6.55	180	6.01	.54
6	2.15	100	2.53	-.38
7	6.60	200	6.88	-.28
8	5.75	160	5.14	.61

$\hat{Y} = -1.82 + .0435X$

Figure: Number of maps distributed and increase in ridership, Transit Example.

Transit example

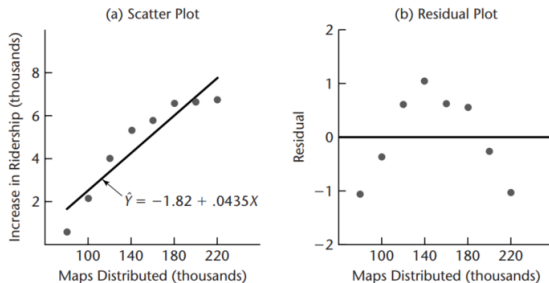
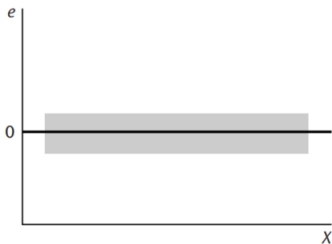
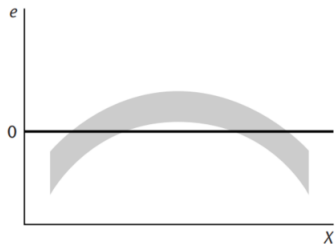


Figure: Number of maps distributed and increase in ridership, Transit Example.

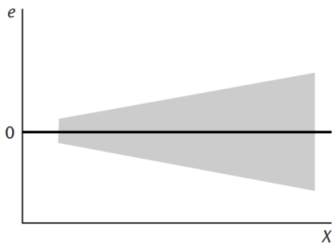
- The residuals depart from 0 in a systematic fashion. Hence, the linear regression model is not appropriate.



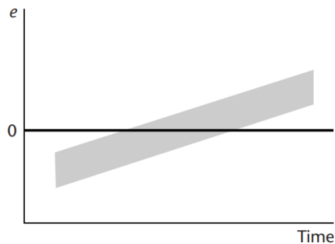
(a)



(b)



(c)



(d)

Figure: Prototype residual plots. Linear regression is appropriate for the data in Fig a, the other cases are not fitted.

Plots of the residuals or of the squared residuals against X or against $X, \hat{Y}$: Nonconstancy of error variance

- Plots of the *residuals* against the predictor variable X or against the fitted values $\hat{Y}$.
- Plots of the *absolute values of the residuals* or of the squared residuals against X or against $\hat{Y}$.

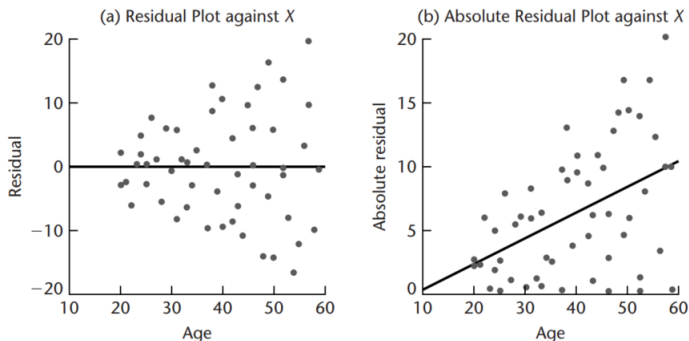
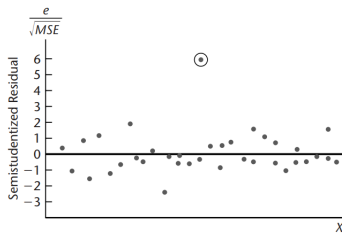


Figure: Example of nonconstant error variance in the study of the relation

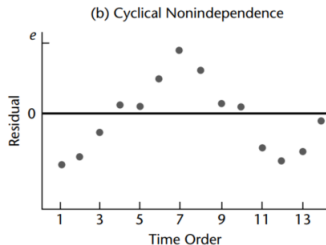
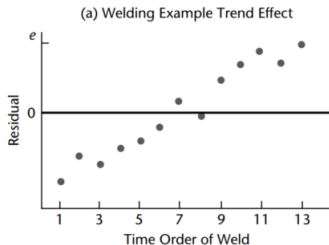
Presence of outliers

- Outliers are extreme observations. Fitted line may be pulled disproportionately toward an outlying observation because outliers causing a misleading fit.
- Can discard the outlier if this observations resulted from a mistake or other extraneous cause.
- On the other hand, outliers may convey significant information, as when an outlier occurs because of an interaction with another predictor variable omitted from the model.
- Plot X vs. semistudentized residual can detect outliers if the dataset is small.



Nonindependence of error terms

- Plot residual against time or in some type of sequence to examine the evident of correlation.



Non-normality of error terms

- The normality of the error terms can be verified by examining the residuals in some of graphic ways.
- Residual is plotted against its expected value (or plot external studentized residuals against the cumulative probability).
- Nearly linear suggests agreement with normality.

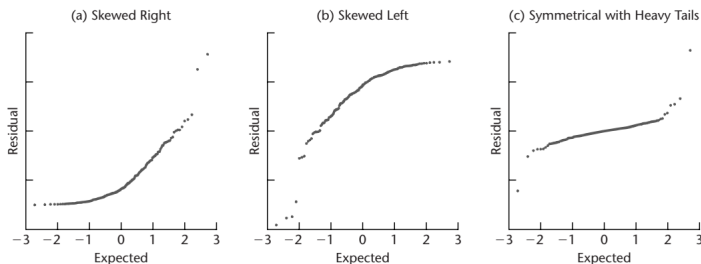


Figure: Normal probability plots when error term distribution is not normal.

Omission of important predictor variables

- Residuals should also be plotted against variables omitted from the model that might have important effects on the response.
- Determine whether there are any other key variables that could provide important additional descriptive and predictive power to the model.

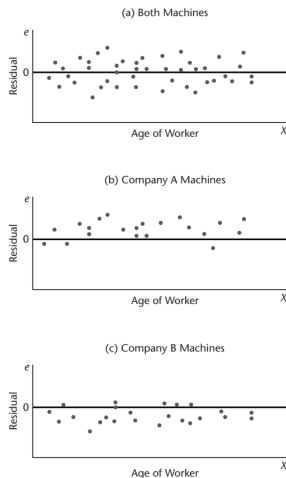


Figure: Residual plots for possible omission of important predictor variable, productivity

example

Residual analysis: Added-variable plots

Residual plots (against X , $\hat{Y}$, Q-Q plot) may not properly show the nature of the *marginal effect of a predictor variable*.

Added-variable plots is used to evaluate this effect.

- **Added-variable plots** is also called **partial regression plots** and **adjusted variable plots**
- Consider the marginal role of a predictor variable X_k , given that the other predictor variables are already in the model.
- Both the response variable Y and the predictor variable X_k are regressed against the other predictor variables and the residuals are obtained for each regression

Added-variable plots

Consider a regression model with two predictor variables X_1 and X_2 . Suppose we are concerned about the nature of the regression effect for X_1 , given that X_2 is already in the model.

- Regressing Y on X_2 obtains the fitted values and residuals

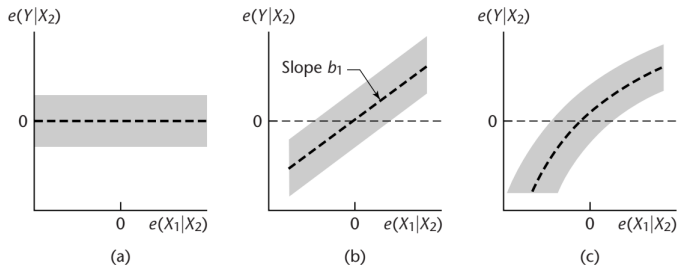
$$\begin{aligned}\hat{Y}_i(X_2) &= b_0 + b_2 X_{i2} \\ e_i(Y|X_2) &= Y_i - \hat{Y}_i(X_2)\end{aligned}$$

- Also regressing X_1 on X_2 yields:

$$\begin{aligned}\hat{X}_{i1}(X_2) &= b_0^* + b_2^* X_{i2} \\ e_i(X_1|X_2) &= X_{i1} - \hat{X}_{i1}(X_2)\end{aligned}$$

- The added-variable plot for predictor variable X_1 consists of a plot of the Y residuals $e(Y|X_2)$ against the X_1 residuals $e(X_1|X_2)$.

Prototype added-variable plots



- Fig.a (a horizontal band): X_1 contains no additional information useful for predicting Y beyond that contained in X_2 , so that it is not helpful to add X_1 to the regression model.
- Fig.b (a linear band): a linear term in X_1 may be a helpful addition to the regression model already containing X_2 . The regression coefficient of X_1 .
- Fig.c (a curvilinear band): the addition of X_1 to the regression model may be helpful and suggesting the possible nature of the curvature effect

Life insurance example

A sample of 18 managers in the 30-39 age group

- X_1 : average annual income of managers during the past two years
- X_2 : manager's risk aversion
- Y : amount of life insurance carried

Manager i	Average Annual Income (thousand dollars) X_{i1}	Risk Aversion Score X_{i2}	Amount of Life Insurance Carried (thousand dollars) Y_i
1	45.010	6	91
2	57.204	4	162
3	26.852	5	11
...	...	...	...
16	46.130	4	91
17	30.366	3	14
18	39.060	5	63

Life insurance example

- Income and risk aversion are mildly correlated here, the coefficient of correlation being $r_{12} = 0.254$
- A fit of the first-order regression model yields:

$$\hat{Y} = -205.72 + 6.2880X_1 + 4.738X_2$$

- Regress Y and X_1 each against X_2 :

$$\hat{Y}(X_2) = 50.70 + 15.54X_2$$

$$\hat{X}_1(X_2) = 40.779 + 1.718X_2$$

- Show student Life Insurance notebook.

Body fat example

Subject	Triceps Skinfold Thickness	Thigh Circumference	Midarm Circumference	Body Fat
i	X_{i1}	X_{i2}	X_{i3}	Y_i
1	19.5	43.1	29.1	11.9
2	24.7	49.8	28.2	22.8
3	30.7	51.9	37.0	18.7
...	...	...	...	...
18	30.2	58.6	24.6	25.4
19	22.7	48.2	27.1	14.8
20	25.2	51.0	27.5	21.1

- X_1 and X_2 are highly correlated ($r_{12} = 0.92$)
- The fitted regression function:

$$\hat{Y} = -19.174 + 0.2224X_1 + 0.6594X_2$$

- Show student the Body Fat notebook.

See some codes at

- https://www.statsmodels.org/stable/generated/statsmodels.graphics.regressionplots.plot_partregress.html
- Regression plots, https://www.statsmodels.org/stable/examples/notebooks/generated/regression_plots.html

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Overview of tests involving residuals

- Tests for randomness: Durbin-Watson test designed for lack of randomness in least squares residuals. This test is discussed when we study autocorrelation.
- Tests for constancy of variance: the Brown-Forsythe test and the Breusch-Pagan test.
- Tests for outliers: discard the suspect outlier and fit a new regression line to the other $n - 1$ observations. This test will be discussed at multiple linear regression.
- Tests for normality: Goodness of fit tests can be used for examining the normality of the error terms. This test will be discussed at multiple linear regression.

Correlation test for normality

- A test for normality of the error terms can be conducted by calculating the coefficient of correlation between the residuals e_i and their expected values under normality.
- Recall that coefficient of correlation $r = \pm\sqrt{R^2} = \sqrt{SSR/SSTO}$. A high value of the correlation coefficient is the indication of normality.
- Toluca company example: the coefficient of correlation between the ordered residuals and their expected values under normality is 0.991. Controlling the α risk at 0.05 for $n = 25$ is 0.959. Hence, we can say that the distribution of the error terms does not depart substantially from a normal distribution.

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Tests for constancy of error variance

- Brown-Forsythe test (reading)
- Breusch-Pagan test

Breusch-Pagan test

- This test assumes that the error terms are independent and normally distributed and that the variance of the error term ϵ_i , denoted by σ_i^2 , is related to the level of X in the following way:

$$\log \sigma_i^2 = \gamma_0 + \gamma_1 X_i$$

- The test $H_0 : \gamma_1 = 0$ versus $H_a : \gamma_1 \neq 0$ is carried out.
- The test statistic X_{BP}^2 is given by:

$$X_{BP}^2 = \frac{SSR^*}{2} \div \left(\frac{SSE}{n} \right)^2,$$

where SSR^* is the regression sum of squares when regressing e^2 on X and SSE is the error sum of squares when regressing Y on X .

- X_{BP}^2 follows approximately the chi-square distribution with one degree of freedom.

Breusch-Pagan test - Toluca company example

- We regress the squared residuals against X and obtain $SSR^* = 7,896,128$.
- We know from previous calculation that $SSE = 54,825$, hence

$$\chi_{BP}^2 = \frac{7,896,128}{2} \div \left(\frac{54,825}{25} \right)^2 = 0.821$$

- To control the α risk at 0.05, we require $\chi^2(0.95; 1) = 3.84$. Since $\chi_{BP}^2 = 0.821 \leq 3.84$, we conclude H_0 , that the error variance is constant.

Summary

- Diagnostics for residuals
- Tests involving residuals

- 1 Chapter 3, Applied Linear Regression Models, Michael H. Kutner *et al.*, 2004
- 2 Chapter 4, Introduction to linear regression analysis, Montgomery, 2014